



Establishing *In-vivo* brain microdialysis for comparing concentrations of a variety of cortical neurotransmitters in the awake rhesus macaque between different cognitive states

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ARTICLE INFO

Keywords:

Brain microdialysis
GABA
Glutamate
Dopamine
Choline
Norepinephrine
Epinephrine
Serotonin
Non-human primates
Behavior
Area MT

ABSTRACT

Background: Neuronal activity is modulated by behavior and cognitive processes. The combination of several neurotransmitter systems, acting directly or indirectly on specific populations of neurons, underlie such modulations. Most studies with non-human primates (NHPs) fail to capture this complexity, partly due to the lack of adequate methods for reliably and simultaneously measuring a broad spectrum of neurotransmitters while the animal engages in behavioral tasks.

New Method: To address this gap, we introduce a novel implementation of brain microdialysis (MD), employing semi-chronically implanted guides and probes in awake, behaving NHPs facilitated by removable insets within a standard recording chamber over extrastriate visual cortex (here, the visual middle temporal area (MT)). This approach allows flexible access to diverse brain regions, including areas deep within the sulcus.

Results: Reliable concentration measurements of GABA, glutamate, norepinephrine, epinephrine, dopamine, serotonin, and choline were achieved from small sample volumes (<20 µl) using ultra-performance liquid chromatography with electrospray ionization-mass spectrometry (UPLC-ESI-MS). Comparing two behavioral states – ‘active’ and ‘inactive’, we observe subtle concentration variations between the two behavioral states and a greater variability of concentrations in the active state. Additionally, we find positively and negatively correlated concentration changes for neurotransmitter pairs between the behavioral states.

Conclusions: Therefore, this MD setup allows insights into the neurochemical dynamics in awake primates, facilitating comprehensive investigations into the roles and the complex interplay of neurotransmitters in cognitive and behavioral functions.

1. Introduction

Cortical neuronal networks, which are underlying various cognitive processes, are modulated by several neurotransmitter systems. The most prominent among these are noradrenergic (norepinephrine – NE and L-epinephrine – LE), cholinergic (acetylcholine – ACh), dopaminergic (dopamine – DA), serotonergic (serotonin – 5-HT), GABAergic (gamma-aminobutyric acid – GABA), and glutamatergic (glutamate – Glu) systems. In primates and rats, these neurotransmitters have been shown to play crucial roles in different aspects of cognition, such as

reward signaling (DA – Anderson et al., 2016; Krebs et al., 2012; Schultz, 2013), working memory and learning (NE – Berridge et al., 2012; Chu and Bloom, 1973; Dalley et al., 2001; De Leo et al., 2023; Mitchell and Weinshenker, 2010; Van Someren, 2021; LE – Gold, 2014; 5-HT – Enge et al., 2011), attentional control (ACh – Cools and Arnsten, 2022; Klínenberg et al., 2011; Roberts et al., 2005; Sarter and Paolone, 2011; DA, NE – McGaughy et al., 2008; Millichap, 1996), and regulation of intrinsic neuronal excitability vs., inhibition for effective information processing at all levels of cognition and behavior (GABA and Glu – Beck and Yaari, 2008; Megías et al., 2001; Petroff, 2002; Pratt and Aizenman,

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<https://doi.org/10.1016/j.jneumeth.2025.110361>

Received 27 August 2024; Received in revised form 2 January 2025; Accepted 4 January 2025

Available online 9 January 2025

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2007; Zhou and Yu, 2018).

Brain microdialysis (MD) is an *in-vivo* sampling technique that concurrently measures the concentration of an extensive range of molecules in a small volume of brain tissue, including various neurotransmitters. This technique, which is typically utilized in freely moving rats (Hermenegildo et al., 2000; Huff and Davies, 2002; Westerink and De Vries, 1988; Young and Bradford, 1986) and less frequently in awake non-human primates (Bradberry et al., 2000; Centeno et al., 2007; Folgering et al., 2019; Thiollier et al., 2018), is often used to observe the effect of both localized and systemic drug administration on brain neurochemistry. A single MD measurement can provide a momentary snapshot of the relative concentration of several different neurotransmitters in the given area of interest. Repeating these measurements over time allows the quantification of dynamic changes in these neurotransmitters, for example, in the context of a drug infusion (for reviews see: Anderzhanova and Wotjak, 2013; Bourne, 2003; Chefer et al., 2009; Darvesh et al., 2011a; Nandi and Lunte, 2009). Furthermore, sampling during various cognitive or behavioral states makes it possible to evaluate the role a specific neurotransmitter or the correlation of different neurotransmitters play in those states. Non-human primates exhibit a more comprehensive array of behaviors than rats and are a better model for human brain physiology. Therefore, examining the underlying neuropharmacological activity in non-human primates is particularly informative for understanding the link between human brain neuropharmacology and cognition/behavior.

A standard MD setup consists of a guide and a probe with a semi-permeable membrane at its tip, placed in the brain area of interest (Fig. 1). This membrane permits the net diffusion of molecules below the molecular weight cutoff of the membrane, from extracellular space (an area of high concentration) to the inside of the probe (an area of low

concentration). The inlet and outlet tubing allows for the perfusion of the probe with artificial cerebrospinal fluid (aCSF) and the collection of samples, respectively.

Prolonged insertion of probes in the tissue can cause gliosis around the probe membrane and increases the risk of inflammation, in turn compromising both the quality and quantity of the samples (for reviews see Watson et al., 2006; Westerink, 1995; original study: Benveniste and Diemer, 1987; further effects of tissue trauma are also discussed in Chen, 2006; Jaquins-Gerstl and Michael, 2009; Nesbitt et al., 2013). Re-inserting removable probes has been observed to trigger only a moderate glial reaction and does not form a glial barrier, leaving the sample output unperturbed (Bradberry et al., 2000; Fumero et al., 1994; Georgieva et al., 1993; Martin-Fardon et al., 1997). Moreover, employing a semi-chronic system enhances the capability for early identification of potential infections on the brain surface area. This is due to the visible access to the dura, which simplifies the cleaning and potential treatment processes. In contrast, chronic implantations that lack visible access to the underlying tissue pose a risk of undetected infections for a long time. Thus, the semi-chronic approach not only mitigates the risk of serious infections but also reduces the necessity for multiple chronic surgeries for the animal.

Microdialysis suffers from poor temporal resolution because of the need for a detectable concentration of the target molecule in the dialysate and, at the same time, the need for a sufficient sample volume for subsequent analysis. This limits the temporal resolution when studying dynamic changes in neurotransmitter concentrations, and it constrains studies that compare different behavioral states of an animal, where the various states need to be maintained for a longer duration.

The limited temporal resolution of MD can be improved by using a particularly sensitive analytical technique for the dialysate. Here, we

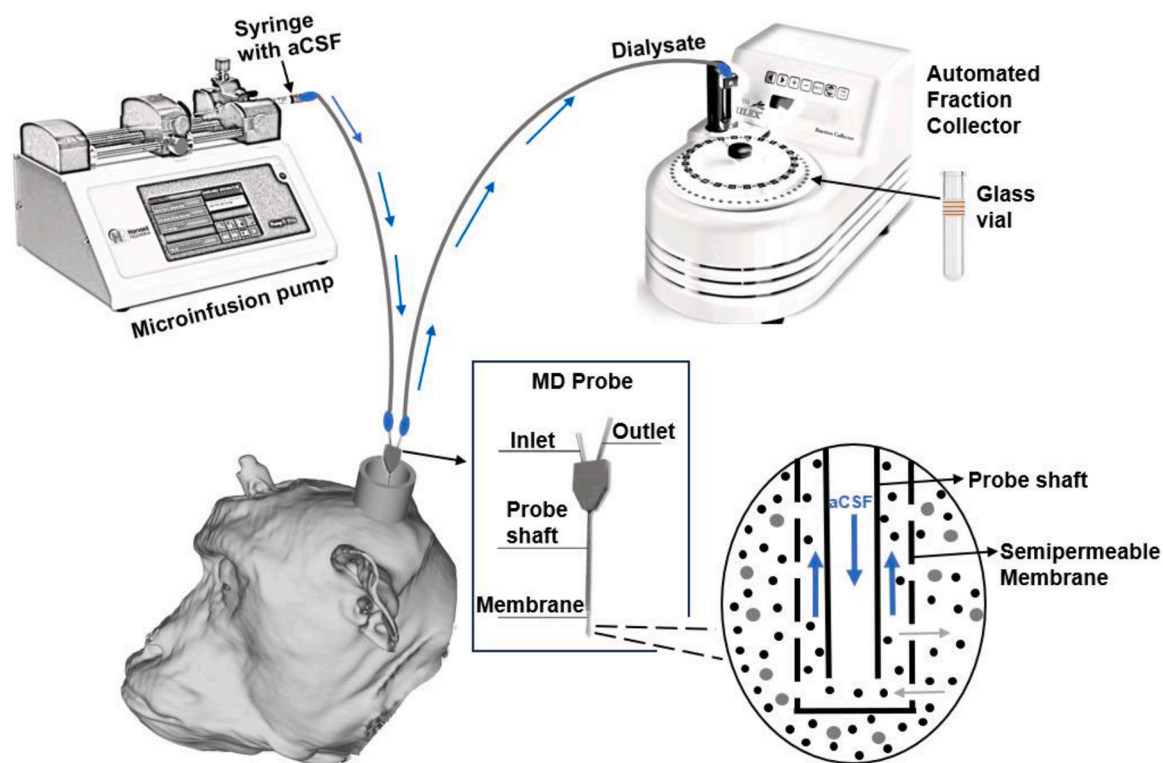


Fig. 1. Schematic of a semi-chronic MD system in rhesus macaques. Note: Before the probe insertion, a guide cannula with a pointy steel guide dummy is inserted into the custom-made inset and carefully lowered inside the chamber. After being positioned, the dummy guide is removed, and the probe is accurately placed into the intended target area through the guide cannula. The system is stabilized inside the recording chamber through a custom-made, 3D-printed cylindrical inset (Fig. 2). The inlet of the probe is connected to a micro infusion pump via the inlet tubing and adapters (blue). The outlet is connected to the automated fraction collector, which collects the dialysate into borosilicate glass vials. Insert: The MD probe incorporates a membrane with a molecular weight cutoff of 20 kDa at its end. The probe facilitates the entry and exit of aCSF passing by the membrane. Due to its semi-permeability, any molecule below the molecular weight cutoff can diffuse into the perfusion stream, enabling a chemical equilibrium between the fluid inside the probe and the surrounding extracellular tissue.

apply Ultra-High Performance Liquid Chromatography-Tandem-Mass Spectrometry (UHPLC-MS/MS) with an electrospray ionization (ESI) step.

In this study, we tackle the following challenges in conducting MD sampling in behaving non-human primates:

- 1) Stable multi-day sampling without glial formation around the MD membrane, which can interfere with the sampling process and may affect the sampling output over time
- 2) Flexibility to access various locations or depths within the target region for high-precision measurements, even within an area located deep in a sulcus, as area MT
- 3) Easy access to the brain surface for routine cleaning, maintaining the integrity of the sampling environment
- 4) Balancing the trade-off between perfusion speed and sampling duration during which the animal performs a behavioral task without compromising the quality of the sample concentration or the animal's behavior
- 5) Robust quantification of nanomolar (nM) concentrations of neurotransmitters in aCSF in a small sample volume (12 μ l)

Research involving non-human primates constitutes a small yet crucial component in neuroscience research. The scientists in this study are aware and are committed to the great responsibility they have in ensuring the best possible science with the least possible harm to the animals (Roelfsema and Treue, 2014; Treue and Lemon, 2023). We have established a comprehensive set of measures to ensure that the severity of our experimental procedures falls into the category of mild to moderate, according to the severity categorization of Annex VIII of the European Union's directive 2010/63/EU on the protection of animals used for scientific purposes (Pfefferle et al., 2018; Cassidy et al., 2024).

2. Materials and methods

2.1. Animal

Information regarding animal care and handling in our lab has been documented earlier (Kozyrev et al., 2019; Schwedhelm et al., 2017; Yao et al., 2016). All animal work was conducted according to the applicable national and international guidelines and German laws governing animal use. All procedures have been approved by the responsible regional government office (Niedersächsisches Landesamt für Verbraucherschutz und Lebensmittelsicherheit [LAVES], Oldenburg, Germany) under permit number 3392 42502-04-13/1100 as well as by the Animal Care Committee of the German Primate Center (DPZ), Göttingen, Germany. All invasive procedures were done under appropriate anesthesia and with appropriate peri-surgical analgesia. The experiments described in this paper were conducted with one adult male rhesus macaque (19 years; 13.5 kg). The animal was group-housed with other macaques in an enriched environment, with different types of toys, wooden structures to climb, and additional enrichment features (Berger et al., 2018; Calapai et al., 2017). It also provides natural and artificial light surpassing regulatory size requirements, ensuring access to outdoor space and facilitating visual and auditory interaction with animals in two adjacent enclosures. During the training and sampling sessions, the animal's primary source of fluid intake was juice rewards, adapted to their preferences. The animal had unlimited access to water and food on days without training or sampling. The German Primate Center employs several veterinarians responsible for monitoring and examining the animals and providing consultation. Throughout this study, the psychological and medical welfare of the animal was continuously monitored by a team of veterinarians, animal facility staff, and the lab's scientists, all of whom were highly experienced in working with non-human primates. The animal previously took part in an optogenetics study. In that study, inhibitory opsins were injected into the right frontal eye field (FEF) of the animal 5.5 years prior to this study,

and we investigated the modulation of neuronal activity in the right area MT with optical stimulation (Hüer et al., 2024). For our current study, we sampled extracellular fluid (ECF) from the left area MT.

2.2. Behavioral task and microdialysis materials

In the current study, semi-chronic MD sampling was done to compare concentrations of six different neurotransmitters and choline in two behavioral states of the animal – 'active' and 'inactive' state. The animal sat in a custom-made primate chair with the head movements restricted. Eye movements were monitored using an eye tracker (EyeLink 1000, SR Research, Ottawa, Canada). The stimuli were presented on a monitor in front of the animal (BenQ XL2720T, resolution 1920 $\times$ 1080, refresh rate 120 Hz). Whenever the animal did not engage in the behavioral task, possibly due to sleepiness or lack of motivation, we consider this an inactive state. In contrast, when the animal engaged in the task, we refer to this as an active state. In that task, we required the monkey to respond to a luminance change of a centrally located red 'fixation square' (0.5 $\times$ 0.5 degrees of visual angle (dva)) while two moving dot patterns of 200 bright dots, moving randomly in a circular aperture (diameter: 5 degrees) flanked the fixation point (Fig. S1; for more detailed information about the task see supplementary section S1). The animal was required to respond within a response time window and received diluted banana or grape juice for every successful response of releasing a touch sensor. The behavioral task was implemented via MWorks (mworks.github.io), an open-source suite of applications and libraries.

To determine the concentration of cortical neurotransmitters, we sampled ECF from area MT located in the posterior bank of the superior temporal sulcus (STS). MD samples were collected from three distinct locations within the area (TR-1, TR-2, TR-3, Fig. 3). In most conventional MD experiments, the guides and probes are chronically inserted in the brain tissue. As described in the introduction, undetected tissue inflammation, neurotransmitter disturbances, and glial formation around the probe membrane are risks associated with such a chronic MD sampling technique. Thus, in the current study, MD guides and probes were implanted semi-chronically using a removable inset that can be easily placed inside an existing recording chamber (Fig. 2). We derived the concept of using removable insets in brain microdialysis from the initial guidance and insights provided by Dr. Corey Roach and Prof. Dr. Anita Disney (Roach, 2023). The 3D-printed inset used in our study (Fig. 2E) is made from biocompatible BioMed clear resin (printed using a Form 3B+ 3D printer by Formlabs, Somerville, MA, USA) and was positioned snugly inside a standard cylindrical polyether ether ketone (PEEK) recording chamber, which was implanted over the target area on the left hemisphere (Additional information on chamber surgery and inset designing is available in the supplementary sections S2 and S3). The inset includes a section ('guide holder') to securely hold the MD guide head, including a circular hole for the actual guide placement, open regions for daily chamber cleaning, and two lateral pins on each side, which prevent rotational or downward movement and ensures a precise alignment with the chamber opening, preventing an upward movement upon chamber closure (Fig. 2C-D).

The MD guide is custom-made from polyurethane with a removable pointy stainless-steel dummy to enable penetration of the dura upon insertion (for detailed information about the MD materials, see supplementary section S4). The MD probe is also custom-made with a stainless-steel shaft. The semipermeable probe membrane is 4 mm long and made of Polyarylethersulfone (PAES) with a molecular weight cut-off of 20 kDa. The stainless-steel shaft provides stability when the probe is lowered into the guide cannula and left in the brain tissue for an extended period.

After lowering the probe into the region of interest, aCSF (final composition: NaCl: 147 mmol/l, KCl: 2.7 mmol/l, CaCl₂: 1.2 mmol/l, MgCl₂: 0.85 mmol/l, Chloride: 153.8 mmol/l), is pumped through the inlet tubing via a 500 μ l Hamilton glass syringe and a dual syringe infusion (Fig. 1). Perfusion is typically conducted at a constant rate of

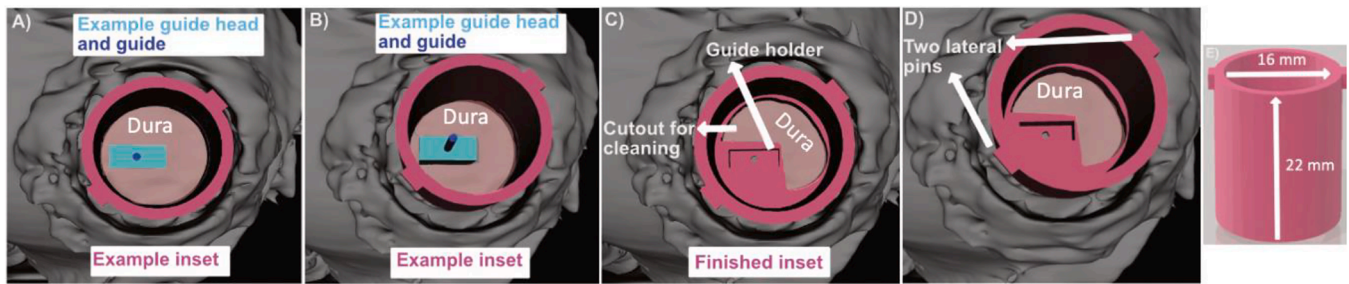


Fig. 2. Inset Design. A) and B) Displays the exemplary inset (pink), including the guide head (light blue) and exemplary guide (dark blue), which are registered onto the previously planned trajectory in the neuronavigation software Cortexplore. The monkey skull and palacos cement fixating the chamber in place over the target area are colored grey. Additionally, the dura is colored light brown. C), D) Illustrate the finished, custom-made, and 3D printed cylindrical inset for semi-chronic MD, with a guide holder including a hole of the diameter of the guide, cutouts for cleaning the chamber, and two lateral pins, securing the inset inside the chamber. E) Displays the finished inset, including inset diameter and height measurements.

0.5–3.0 $\mu\text{L}/\text{min}$, ensuring an ideal diffusion rate driven by a concentration gradient between the perfusion fluid in the probe and the extracellular space (Anderzhanova and Wotjak, 2013; Chefer et al., 2009; Westerink, 1995). Decreasing the flow rate is advisable, particularly for analytes with low extracellular concentration or diffusivity, thereby enhancing the likelihood of obtaining a measurable analyte concentration in each sample (Chefer et al., 2009). Low flow rates extend the collection times, rendering them unsuitable for most experiments involving behavioral tasks with awake, behaving animals. Therefore, we selected a constant flow rate of 1 $\mu\text{L}/\text{min}$ (1 $\mu\text{L}/\text{min}$; 12 min sampling = 12 μL / sample) to optimize the trade-off between time resolution and analyte recovery.

The inlet and outlet tubing are made from Fluorinated ethylene propylene (FEP) and are each 1 m long, with an inner diameter of 0.12 mm, and connected to the pump, as well as the autosampler via zero internal volume tubing adapters (Fig. 1). The samples are stored in borosilicate glass vials, containing a 20:1 ratio of sample to concentrated acetic acid (0.6 μL of acetic acid for 12 μL sample) to prevent oxidation of catecholamines (Chefer et al., 2009). After that, the vials are placed in the autosampler, which keeps the samples refrigerated at 4 $^{\circ}\text{C}$ and collects them every 12 minutes (the time it takes for the dead volume to pass through the inlet and outlet tubing). This enables automatic MD sampling without disturbing the animal in several behavioral blocks of 12 minutes each. Immediately after collecting a sample, it is transferred to an ice box containing dry ice and later stored at -80°C until UHPLC-ESI-MS analysis.

2.3. Planning trajectories for the inset positioning

Samples were collected from three distinct locations within the target region (Fig. 3A). The trajectory planning was conducted using a neuronavigation software and hardware suite called Cortexplore (cortexplore GmbH, Linz, Austria; for more details, see (Crayen et al., 2024)). The tool utilizes MRI and CT scans to select a target location in the region of interest and a location on the surface of the dura where the probe will enter. The line between these two points is a ‘target trajectory.’ Three such trajectories were planned before the placement of the recording chamber (Fig. 3A). After the chamber placement, the trajectories were revised to calibrate them to the actual placement of the chamber and ensure they were parallel to the chamber’s trajectory (Fig. 3C, D, and E). This was important, as this minimized tissue damage while lowering the guide, which was advanced along the chamber’s trajectory. A separate inset was designed for each trajectory, featuring a hole for the guide and a guide-holding mechanism centered around the respective target trajectory. Please refer to the supplementary section S3 for stepwise instructions on trajectory planning for inset design.

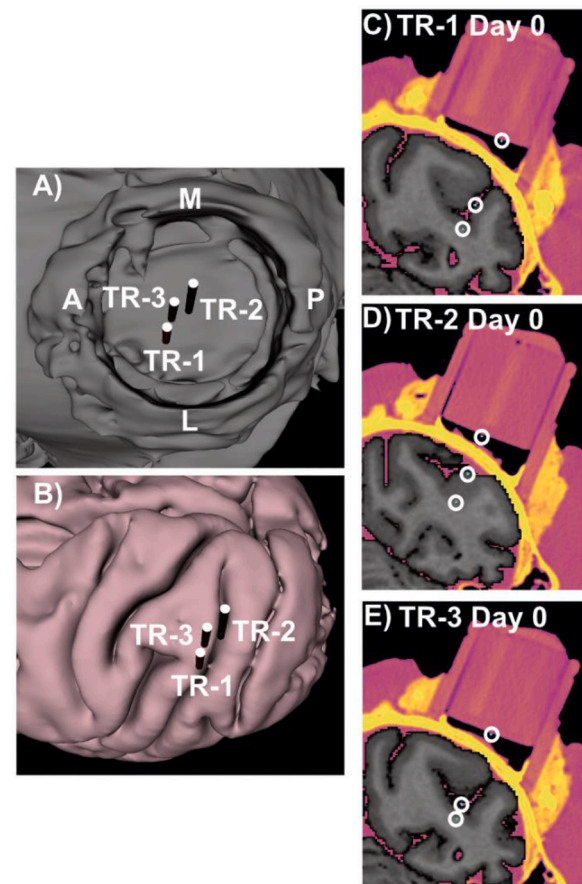


Fig. 3. The three MT trajectories used for MD sampling (TR-1, TR-2, and TR-3; TR=trajectory). A) Displays the three MT trajectories on the skull model (grey) inside the chamber outline (grey, circular elevation). B) Shows the brain model and the location of the three MT trajectories used for MD sampling. C)-E) Illustrate the chamber outline (over the left hemisphere), the dummy inset used during the CT scan (pink rectangular object), and the planned locations of the MD guide shaft within the inset. The three white circles display the three target points used for planning. Starting at the inset with increasing penetration depth: the location of the probe shaft in the inset (point 1), the end of the guide/beginning of the membrane (point 2), and the end of the membrane in the target location (point 3). C) Illustrates the planned MD trajectory of TR-1 on Day 0 (before any MD sampling) in the coronal plane (left hemisphere), D) Displays the planned MD trajectory of TR-2 on Day 0 (before any MD sampling) in the coronal plane (left hemisphere) E) Shows the planned MD trajectory of TR-3 on Day 0 (before any MD sampling) in the coronal plane (left hemisphere).

2.4. Microdialysis procedure

Each MD data collection cycle consists of one day of animal training and six days of MD sampling. Each cycle uses a new trajectory. On the training day of each cycle, an inset (Fig. 2), along with the polyurethane guide, including a stainless-steel pointy guide dummy, is placed inside the chamber to enable the piercing of the dura. After 5 minutes, the stainless-steel dummy is carefully removed while the guide stays in place, and the probe is inserted into the guide. The chamber is cleaned and sealed with a chamber cap. No samples are collected on the training day to ensure a recovery time of the tissue for at least 18 hours. This is done to allow for settling of trauma responses around the tissue due to the insertion, to accommodate for inflammatory tissue responses with respect to changes in the local glucose metabolism, blood flow and ensure closing of the blood-brain barrier, as well as accounting for neurotransmitter disturbances (for reviews see: [Anderzhanova and Wotjak, 2013](#); [Beneviste, 1989](#); [Watson et al., 2006](#); [Westerink, 1995](#)).

On sampling days, the collection vials are filled with acetic acid, labeled with the corresponding task block identifier (based on a pre-assigned random order), and inserted into the autosampler, which refrigerates the samples at 4 °C (Fig. 1). The glass syringe is filled with 500 μ L of sterile aCSF and placed in the infusion pump. To test the tubing and the adapter for small holes and remove the air bubbles, the inlet tube and outlet tube are connected via a tubing adapter. The inlet tube is then attached to the glass syringe. The aCSF is pumped through at a flow rate of 5 μ L/min until all air bubbles are removed. Afterwards, one end of the inlet tube is left connected to the pump syringe, while the other is connected to the probe's inlet. The outlet tube, on the other hand, is connected to the outlet of the probe and the autosampler with additional tubing adapters (Fig. 1). Samples are collected at a flow rate of 1 μ L/min, collecting 12 μ L of the sample during each 12-minute block.

During the initial three "stabilization blocks," the animal is not provided with a task, allowing for visual observation of the proper output sample volume in the vial (Fig. 4) without disturbing the animal's task performance due to entering the training room. If the output is different than 12 μ L, any flow obstruction is removed, the flow rate is rechecked, and the session is restarted. After the stabilization blocks, the animal is presented with task blocks in a randomized order (Fig. 4). At the end of the session, the vials are immediately moved from the autosampler to dry ice and transferred to a -80 °C storage within hours. After removing the vials, the inlet and outlet tubes are carefully disconnected from the

probe, and the chamber is cleaned. The inset containing the guide and probe is left inside the tissue overnight, and on the following days, more MD sessions can be performed.

The guide is left in place for the whole time of one MD sampling cycle, while the probe is exchanged at the end of the third day's sampling session and replaced by a fresh probe, which is used for the remaining three days of MD sampling. This is done to account for potential gliosis and clogging of the membrane. It has been shown before that, once inserted, a probe can be left in the tissue for about 72 hours without causing any alterations to the sample recovery ([Zamboni, 2007](#)).

On the seventh day, the entire apparatus is removed from the chamber. Afterwards, the animal can recover and access water ad libitum for the subsequent two days until a new MD sampling cycle is started. In this study, we conducted three cycles, resulting in 18 MD sampling days (sessions). We collected 9–12 samples (each: 12 mins collection) each session.

2.5. Confirmation of probe location

To confirm the location of the MD probe within the target region, we conducted a post-dialysis MRI scan, which was aligned with the pre-dialysis MRI scan in Cortexplore. The post-dialysis MRI revealed that the MD guide and probe were placed in our target region, and MD samples were indeed collected from area MT (for reference, see supplementary section S6 and Fig. S2). A more conventional alternative would be a post-mortem immunohistological verification. While this offers potentially higher accuracy than the MRI-based approach, it prevents further data collection and is preferable from the point of the 3 R (replacement, reduction, and refinement) principle.

2.6. Ultra-Performance Liquid Chromatography-Electrospray Ionization-Mass Spectrometry (UHPLC-ESI-MS/MS)

The flow rate in microdialysis creates a particular challenge in behavioral tasks, as it determines how long a particular task needs to be performed and limits the possibility to alternate behavioral conditions. Thus, this study design poses an analytical and technical challenge due to two factors: 1) the limited quantities of neurotransmitters collected (i. e., nM range) when sampling during endogenous, physiological states and 2) a small MD sample volume due to the low flow rate. Additionally, MD brain samples contain a mixed number of inorganic salts from the

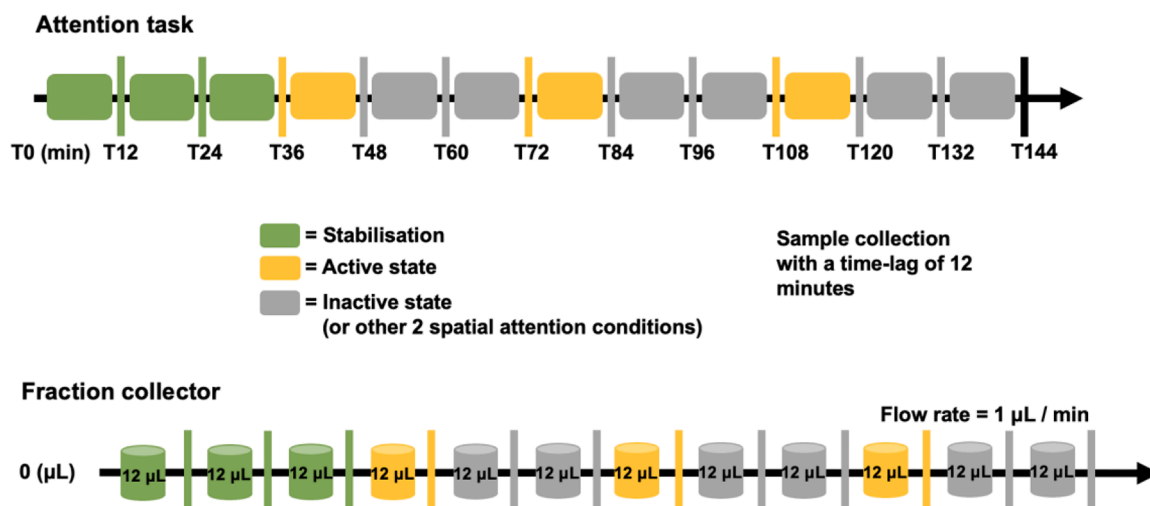


Fig. 4. Temporal dynamics of an MD experimental session. The experiment is started with three stabilization blocks of 12 minutes each, during which the outflow is routinely monitored (1 μ L/min, total of 12 μ L after 12 minutes). Following the stabilization period, the animal engaged in a behavioral task ("active" state, yellow) in blocks of 12 minutes. Blocks where the animal did not engage in the task are referred to as the "inactive State" (grey). Due to the outlet dead volume, the sample collection is shifted by 12 minutes from the behavioral task block, shown on the screen.

perfusion fluid (aCSF). Thus, chromatographic separation is necessary to separate and subsequently detect the different neurotransmitter molecules (Zhang and Beyer, 2006).

The relative concentration of six neurotransmitters and choline was quantified at Biofidus AG (Bielefeld, Germany) after performing a feasibility study (for more information see supplementary section S5). Before each measurement, each sample vial is prepared by diluting the sample volume 1:2 in 100 % aCSF with 0.1 % formic acid (FA) (v/v). The compounds are separated on an UHPLC system (Agilent Infinity 1290 II), using a reversed-phase column (AcclaimTM C30, 3 μ m, 2.1 $\times$ 150 mm, Dionex Bonded Silica Products, Thermo Fisher Scientific). Mobile phases are 0.1 % FA in water (A) and acetonitrile (B). The mass spectrometric analysis is performed with an Ultivo triple quadrupole QQQ mass spectrometer (Agilent) using the dynamic Multiple Reaction Monitoring (MRM) mode (ion source parameters: capillary voltage 2500 V, gas flow 10 L/min, gas temperature 300 °C, Nebulizer 40 psi, Nozzle voltage 0 V, sheath gas flow 12 L/min, the sheath gas temperature 350 °C; profile of the gradient: 0.1 min 10 % B, 1.2 min

20 % B, 2 min 30 % B, 2.4 min 95 % B, 4.2 min 95 % B, 4.6 min 0 % B, 5.0 min 5 % B, 5.4 min 100 % B, 11.4 min 100 % B; MRM parameter see Table S1).

3. Results

The UHPLC-ESI-MS approach quantified the relative concentrations of six neurotransmitters (DA, NE, LE, 5-HT, Glu, GABA) and choline, which is used as a proxy for acetylcholine measurements, in low nanomolar ranges (>0.5 nM) using low sample volumes (12 μ l per sample) in 18 samples (9 active and 9 inactive; collected across 18 days). To control for type 1 error, which can happen in samples with small compound composition, measurements were taken at three separate time points, resulting in triplicate measurements for each compound (mean values for each triplicate, the standard error, and percentage error are detailed in Table S5). For GABA, 13 out of 18 samples had percentage errors below 8 %, while 5 samples ranged between 16.9 % and 29.6 %. For glutamate, 17 out of 18 samples had errors below 4 %, with only one

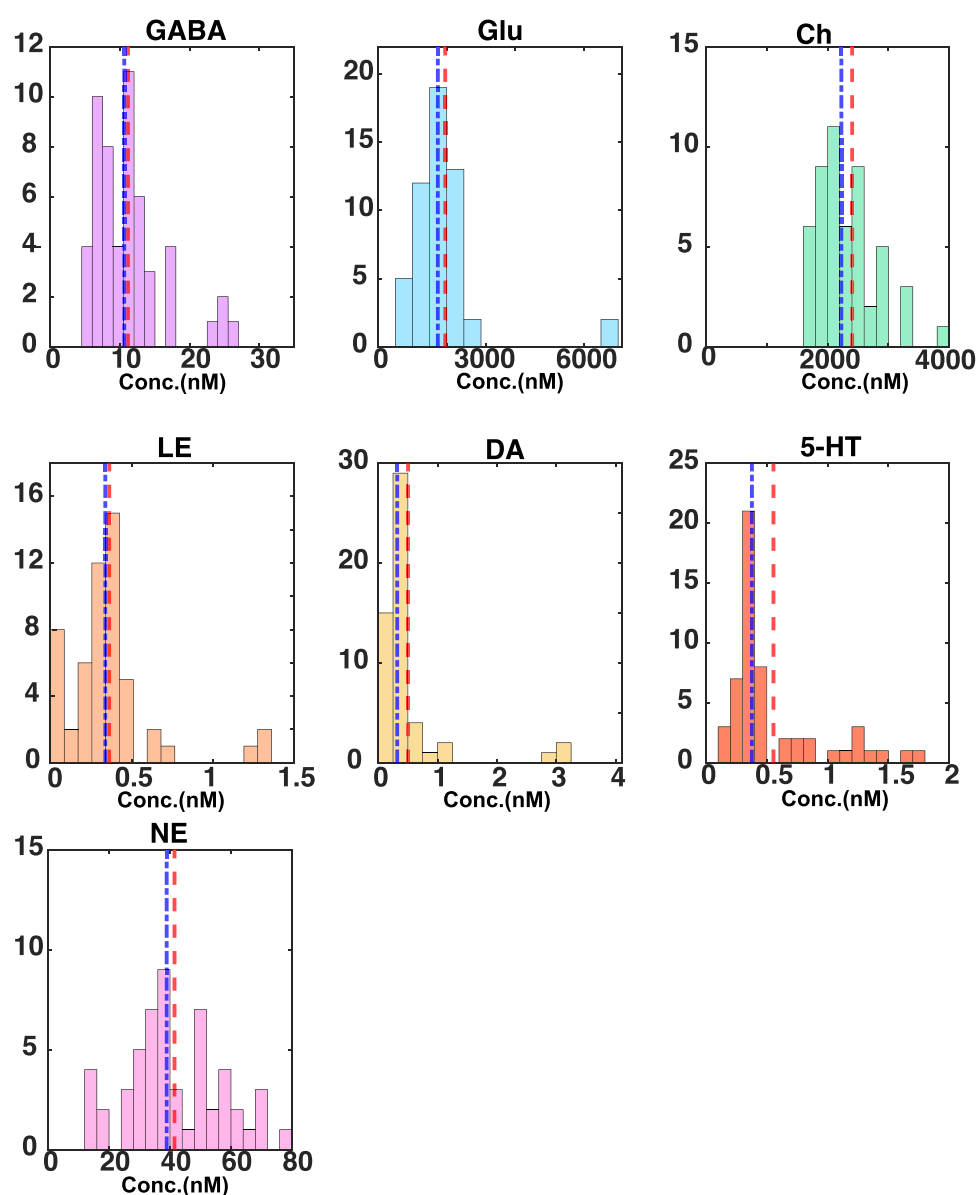


Fig. 5. Frequency distributions of concentrations of seven neurotransmitters. The red dashed line indicates the mean concentration of the distribution and the blue dashed line indicate the median. $N = 54$. Raw triplicate concentration values are provided in [supplementary table S3](#). GABA; Glutamate: Glu; Choline: Ch; Norepinephrine: NE; Epinephrine: LE; Dopamine: DA; Serotonin: 5-HT.

sample at 21.5 %. Choline had errors below 2 % for all samples. Norepinephrine showed errors below 5 % in 15 samples and below 8 % for the rest. For Epinephrine, 8 samples had errors below 8 %, 5 below 20 %, and 5 below 60 %. Dopamine had errors below 10 % in 6 samples, below 20 % in 11 samples, and one sample had an error above 20 %. For serotonin, 14 samples had errors below 10 %, while the rest was below 22 %. Although some of the samples showed variability in their percentage errors, most did not exceed beyond 20 %. For samples above the 20 % error threshold, it is hard to determine whether that is due to the instrument error or due to inherent variability in compound concentration resulting from a secondary factor that we cannot control. [Supplementary table S3](#) furthermore depicts the raw concentration values per sample (triplicate).

We further validate our methodological approach for sampling extracellular fluid and detecting the presence of the neurotransmitters in macaque area MT by subjecting the obtained concentration values to additional analysis. [Fig. 5](#) shows the frequency histograms of the concentrations for GABA, glutamate, choline, norepinephrine, epinephrine, dopamine, and serotonin pooled across ‘active’ and ‘inactive’ states. The mean and median values are marked with red and blue dashed lines, respectively – for GABA (mean: 11.1 nM, median: 10.6 nM), glutamate (mean: 1960 median: 1740 nM), choline (mean: 2400 nM, median: 2230 nM), norepinephrine (mean: 41.4 nM, median: 38.9 nM), epinephrine (mean: 0.362 nM, median: 0.338 nM), dopamine (mean: 0.503 nM, median: 0.320 nM), and serotonin (mean: 0.555 nM, median: 0.378 nM). Note that these concentrations reflect the concentration in the sample, *not* the actual concentrations in the tissue surrounding the probe. This aspect of microdialysis sampling makes a direct comparison of concentrations between studies challenging (see discussion for more). We still find these concentration values in the same ranges as most of the values found in other studies (see [supplementary table S2](#)).

[Fig. 6](#) presents the distribution of concentration values across two behavioral states—active and inactive- to validate the concentration values in the context of behavioral manipulation. For most neurotransmitters, we only find minor differences between the neurotransmitter concentrations in the two behavioral states. A Wilcoxon rank sum test

indicates, though, that the concentration of GABA during the active state (median: 11.5 nM) is significantly higher than during the inactive state (median: 8.31 nM; $z = 2.2$, $p = 0.02$). Similarly, the serotonin concentration in the active state (median: 0.478 nM) is significantly higher than in the inactive state (median: 0.348 nM; $z = 2.8$, $p = 0.003$). All mean and median values for the six neurotransmitters and choline in each of the two behavioral states are depicted in [table S4](#). To further assess the variability of concentration measures in the two behavioral states, we computed the coefficient of variability (CV)—a dimensionless measure of dispersion around the mean, suitable for comparing datasets with varying means—for each neurotransmitter across each behavioral state. CV values were consistently higher in the active state than the inactive state ([Fig. 6B](#), last “variability” plot), possibly reflecting greater variability in behavioral performance during the active state versus the stable, low-arousal state of the inactive condition. [6B](#), ‘variability’ plot), possibly reflecting greater variability in behavioral performance during the active state versus the stable, low-arousal state of the inactive condition.

To better understand the *complex interplay* between neurotransmitters, we investigated how their concentrations fluctuate relative to each other within a sample. By computing Spearman’s correlations for each neurotransmitter pair, we gained insights into their co-regulation and dynamic interactions across the two behavioral conditions ([Fig. 7A](#) for the “active” state; [Fig. 7B](#) for the “inactive” state). Significant correlations indicate a link between the two neurotransmitters. In the active state, significant positive correlations were found between excitatory neurotransmitters such as dopamine, choline, norepinephrine, glutamate, and the inhibitory neurotransmitter, serotonin ([Fig. 7A](#), orange-colored data sets). In the inactive state, GABA and epinephrine showed significant correlations with other neurotransmitters, with GABA displaying strong positive correlations with choline and epinephrine and a strong negative correlation with norepinephrine ([Fig. 7B](#), orange-colored data sets).

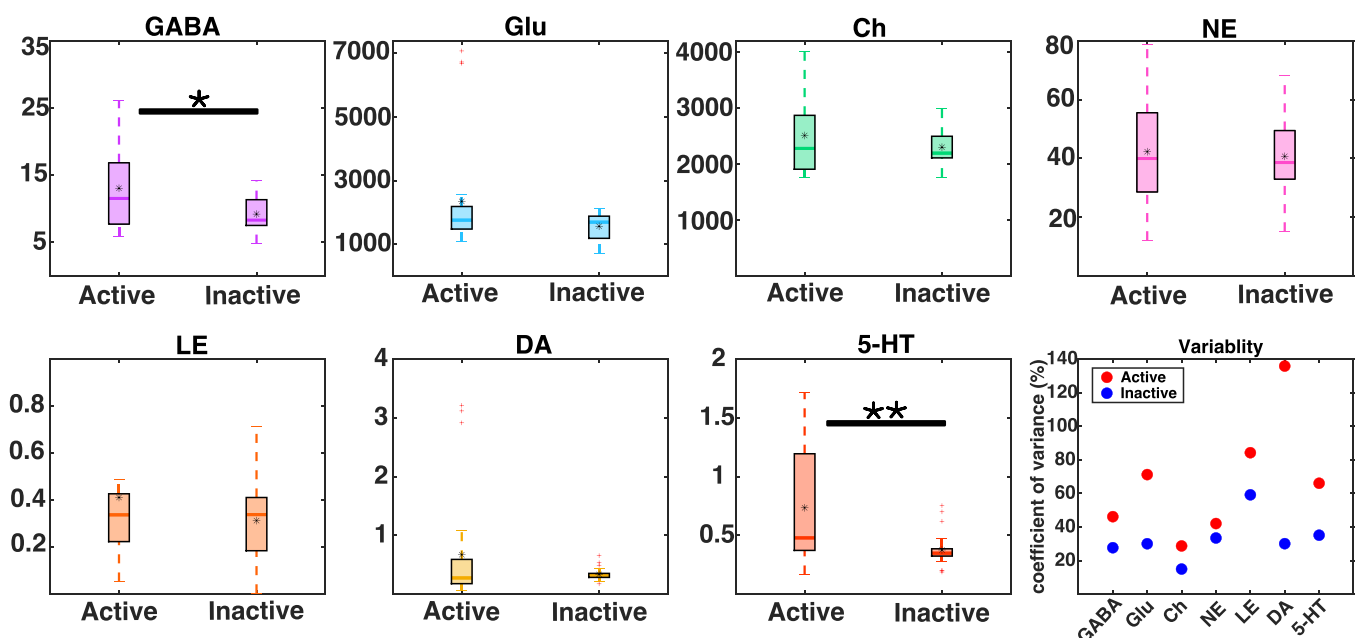


Fig. 6. Boxplots depicting the concentration values of the seven neurotransmitters in the samples from ‘active’ and ‘inactive’ behavioral blocks and the variability (CV) measure for the two behavioral blocks. Significant differences were only found between the two blocks for the concentrations of GABA and serotonin in the sample. **Red crosses** indicate the outliers in the data, **small black points** indicate the mean concentration values and the **solid line** depicts the median. The CV values for active blocks (red) are consistently higher for inactive blocks (blue), reflecting higher variability in behavioral performance. GABA; Glutamate; Glu; Choline; Ch; Norepinephrine; NE; Epinephrine; LE; Dopamine; DA; Serotonin; 5-HT.

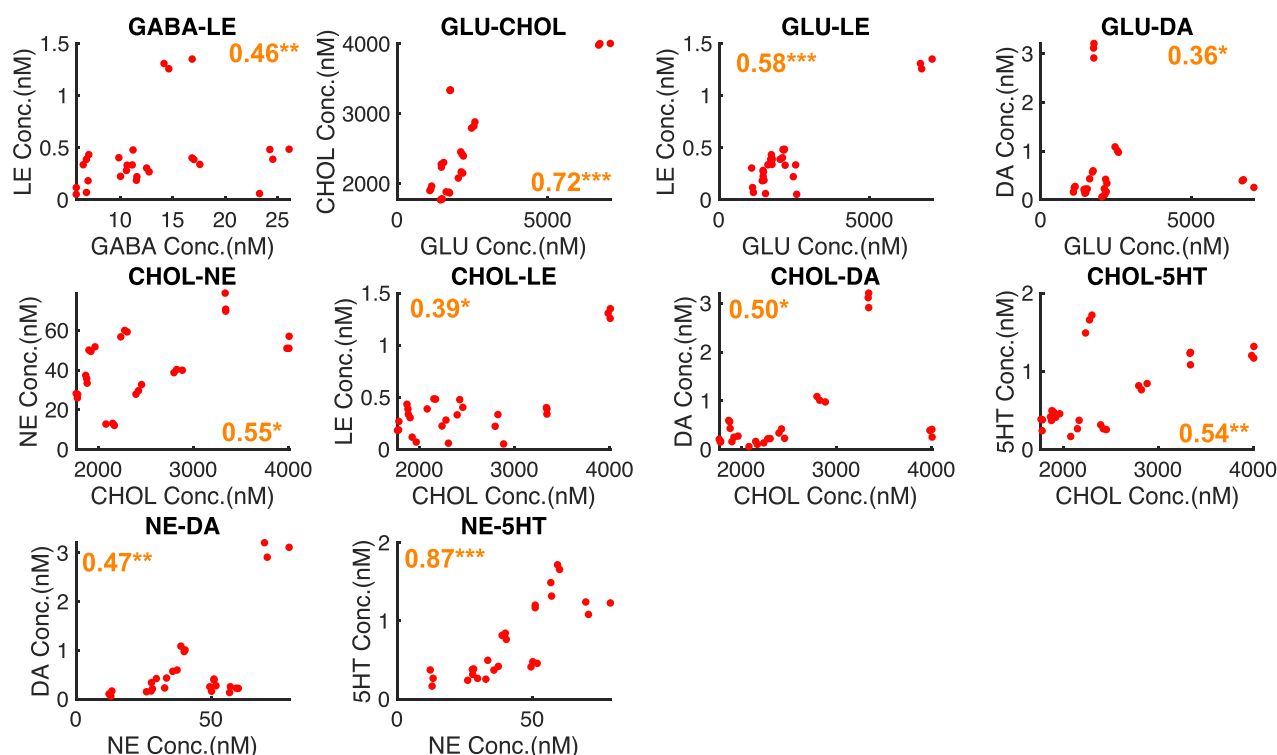
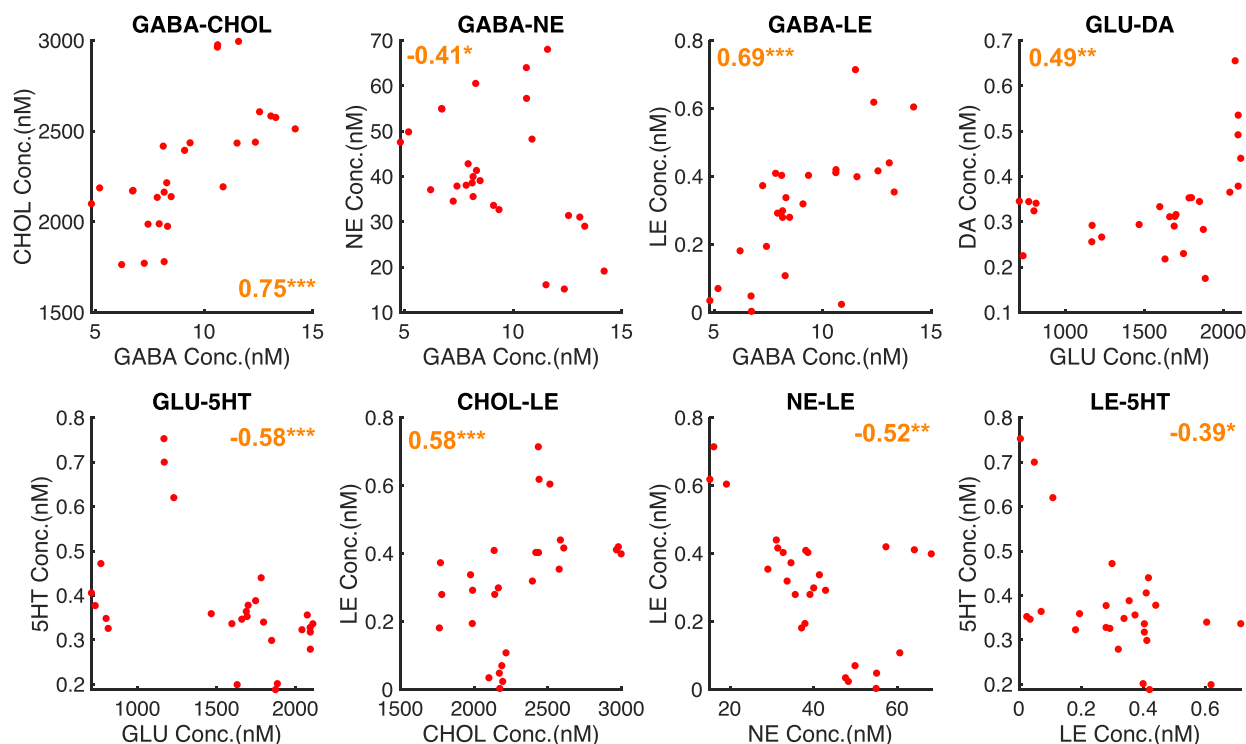
A**Correlations between neurotransmitters during "active" behavioral state****B****Correlations between neurotransmitters during "inactive" behavioral state**

Fig. 7. Spearman's correlations in 'active' and 'inactive' states. The positive coefficient entails a positive relation between the two concentrations. The strength of the correlations and the statistical significance are indicated in red.

4. Discussion

In this study, we collected extracellular fluid from the visual cortical area MT of a behaving rhesus monkey via semi-chronic microdialysis. We designed a removable 'inset' allowing the system to be adapted to any MD guide, probe, and recording chamber configuration without subjecting the animal to multiple surgeries to reach the desired brain region. Our semi-chronic approach provides a high degree of flexibility in sampling multiple locations and depths within a region of interest while providing the same advantages of conventional chronic MD, such as stability of guide and probe in the tissue and continuous sampling over multiple days. Additionally, with the semi-chronic approach, we were able to easily access the dura, enabling the maintenance of the integrity of the tissue over an extended period. We successfully demonstrate that MD can be used to determine the relative concentration between two behavioral states of a variety of neurotransmitters in low nM ranges from awake and behaving nonhuman primates.

We employed UHPLC-ESI-MS to reliably measure six neurotransmitters: GABA, glutamate, norepinephrine, epinephrine, dopamine, serotonin, as well as choline in a small sample volume (12 μ l) with concentration values in low nanomolar ranges (Fig. 5) comparable to what was reported previously in the literature (supplementary table S2). As the total sample output during MD relies on the perfusion flow rate and usually underestimates the basal extracellular concentrations, it has been emphasized that MD is a relative sampling technique in which the results should be expressed and interpreted as relative changes in neurotransmitter concentrations within the dialysate, elucidating the averaged temporal fluctuations during, e.g., behavioral changes (Di Chiara et al., 1996; Westerink, 1995). This is one of the aspects of microdialysis sampling that makes a direct comparison of concentrations between studies challenging.

To classify our results in the context of results from other studies, we still decided to look at the concentrations reported in other studies (supplementary table S2). The distributions for concentrations of GABA, epinephrine, glutamate, serotonin, and dopamine (Fig. 5) are comparable to those reported by several other studies in rats (supplementary table S2). One exception is the concentration of norepinephrine, which was substantially higher than the values reported for rats. This could possibly reflect a species difference. In UHPLC-MS analysis, ACh and choline often require separate analyses due to their different chemical structures and properties, affecting their behavior in chromatography and subsequent detection. Due to the low sample volume obtained in this study, a separate analysis of ACh and choline was not possible here. As choline can be analyzed with all the other compounds of interest, we decided to analyze choline instead of ACh. However, it should be noted that not all choline comes from the ACh breakdown, so choline and Acetylcholine concentrations do not directly correlate with each other (Blusztajn and Wurtman, 1983; Darvesh et al., 2011b).

Additionally, the concentration of epinephrine was obtained opportunistically, given the technical capabilities of UPLC-ESI-MS.

Additionally, we measured relative neurotransmitter concentration changes (in 18 samples collected over 18 MD sessions) associated with endogenous physiological fluctuations caused by differing cognitive states. The basal concentration of neurotransmitters may fluctuate on a day-to-day basis. It is therefore helpful to collect baseline samples, to use as a reference for computing task-based neurotransmitter fluctuations. Our comparison between the animal's 'active' and 'inactive' states showed only minor differences in neurotransmitter concentrations between the two states (Fig. 6a). However, the active state exhibited greater variability in concentrations, likely due to fluctuating behavioral performance during this phase. Each 12-minute block of activity allowed the animal to engage and disengage with the task intermittently, contributing to the variability of neurotransmitter concentrations between the nine samples of the active state (Fig. 6). In contrast, the consistently low variability of concentrations from the inactive state, collected when the monkey was resting or sleeping, might reflect a

stable low arousal condition.

Next, we looked at how the concentrations of these neurotransmitters fluctuate with respect to each other within the same behavioral state to investigate neurotransmitter dynamics and interactions. During the active state, neurotransmitters such as dopamine, choline, norepinephrine, glutamate, and serotonin display significant positive correlations, suggesting joint dependence on engagement and task performance (Fig. 7a). This may reflect that the task requires a certain balance of excitation, inhibition and different neurotransmitter influences for the monkey to perform the detection of luminance change. In contrast, GABA and epinephrine exhibit significant correlations with other neurotransmitters during the inactive state, with GABA showing a strong positive correlation with choline and epinephrine and a negative correlation with norepinephrine. This may reflect a strong GABA-induced suppressive modulation during the inactive state. This prominent state-dependency of the correlation of GABA concentrations with other neurotransmitters may reflect the importance of the excitatory-inhibitory balance underlying modulations in neuronal activity (Petroff, 2002). Consequently, this underscores the premise that complex brain functions are not governed by one neurotransmitter but rather through the joint actions of multiple chemical messengers.

The observation of neurotransmitter concentrations comparable to what has been reported before (supplementary table S2) supports the reliability of the technique. Similarly, we find differences in neurotransmitter concentrations and variabilities between behavioral states, in line with the hypothesis of a causal role of neurotransmitters in the state-modulation of brain activity (Aston-Jones and Bloom, 1981; Harris and Thiele, 2011; Lee and Dan, 2012; McGinley et al., 2015; Schuerman et al., 2022; Van den Brink et al., 2019). Furthermore, we demonstrate that with this technique, we can characterize correlations between different neurotransmitters which have been suggested to have joint actions in the brain (Alex and Pehek, 2007; Blier, 2001; Di Giovanni et al., 2008; Harris and Thiele, 2011; Lee and Dan, 2012; Pisani et al., 2000; Van den Brink et al., 2019; Weiss et al., 1981).

4.1. Conclusion

In summary, we devised a novel MD setup utilizing semi-chronically implanted MD guides and probes for multi-day sampling of extracellular fluid in awake and behaving non-human primates. Here, we highlight some of the unique features of this method:

- **Semi-chronic microdialysis via removable insets using a standard electrophysiology implant**
 - o Removable insets can be easily implemented in combination with other methodologies, such as electrophysiology and microinjections, without the need for specialized MD implants (Bradberry et al., 2000).
 - o Removable insets facilitate implant cleaning.
 - o Removable insets allow for flexibility in terms of recording location and probe angle.
- **Reliable, simultaneous measurements of multiple neurotransmitters in a small sample volume from macaque sensory cortex.**
- **Quantification of multiple neurotransmitters simultaneously during different cognitive states.**

To conclude, systems neuroscience research has focused on electrophysiological recordings of neural activity and its modulation when investigating cortical information processing. Due to a lack of suitable and accessible methods, very few studies have looked at the functional role of neurotransmitters, especially when investigating the mechanisms underlying cognitive neural activity modulations. The semi-chronic microdialysis employed here measures relative concentrations of multiple neurotransmitters in parallel under specified behavioral conditions and can potentially be expanded to sample from multiple brain regions

simultaneously.

Funding sources

This project is supported by a grant to Stefan Treue from the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation, <http://www.dfg.de/en>) – project number 436260547.

CRediT authorship contribution statement

Stefan Treue: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Max Arwed Crayen:** Writing – review & editing, Validation, Software, Methodology. **Pankhuri Saxena:** Writing – review & editing, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Stella Mayer:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization.

Declaration of Competing Interest

Authors report no conflicts of interest.

Acknowledgements

The authors are very grateful for the expert help by the veterinary and animal husbandry staff of the DPZ for animal care and to Leonore Burchardt, Sina Plümer, Klaus Heisig, and Ralf Brockhausen for exceptional technical and animal-related assistance. Special thanks to Dr. Tobias Lutterman from Biofidus A.G. for conducting UHPLC-ESI-MS on our samples. Additionally, we would like to express our profound gratitude to Prof. Dr. Anita Disney and Dr. Corey Roach for their invaluable insights into their MD approach, being instrumental in setting up our MD setup. We are deeply grateful for their support and generosity.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jneumeth.2025.110361](https://doi.org/10.1016/j.jneumeth.2025.110361).

Data availability

Data provided in the supplementary material.

References

- Alex, K.D., Pehek, E.A., 2007. Pharmacologic mechanisms of serotonergic regulation of dopamine neurotransmission. *Pharmacol. Ther.* 113 (2), 296–320. <https://doi.org/10.1016/j.pharmthera.2006.08.004>.
- Anderson, B.A., Kuwabara, H., Wong, D.F., Gean, E.G., Rahmim, A., Brašić, J.R., George, N., Frolov, B., Courtney, S.M., Yantis, S., 2016. The role of dopamine in value-based attentional orienting. *Curr. Biol.* 26 (4), 550–555. <https://doi.org/10.1016/j.cub.2015.12.062>.
- Anderzhanova, E., Wotjak, C.T., 2013. Brain microdialysis and its applications in experimental neurochemistry. *Cell Tissue Res.* 354 (1), 27–39. <https://doi.org/10.1007/s00441-013-1709-4>.
- Aston-Jones, G., Bloom, F.E., 1981. Activity of norepinephrine-containing locus coeruleus neurons in behaving rats anticipates fluctuations in the sleep-waking cycle. *J. Neurosci.* 1 (8), 876–886. <https://doi.org/10.1523/JNEUROSCI.01-08-00876.1981>.
- Beck, H., Yaari, Y., 2008. Plasticity of intrinsic neuronal properties in CNS disorders. *Nat. Rev. Neurosci.* 9 (5), 357–369. <https://doi.org/10.1038/nrn2371>.
- Benevise, H., 1989. Brain Microdialysis. *J. Neurochem.* 52 (6), 1667–1679. <https://doi.org/10.1111/j.1471-4159.1989.tb07243.x>.
- Benveniste, H., Diemer, N.H., 1987. Cellular reactions to implantation of a microdialysis tube in the rat hippocampus. *Acta Neuropathol.* 74 (3), 234–238. <https://doi.org/10.1007/BF00688186>.
- Berger, M., Calapai, A., Stephan, V., Niessing, M., Burchardt, L., Gail, A., Treue, S., 2018. Standardized automated training of rhesus monkeys for neuroscience research in their housing environment. *J. Neurophysiol.* 119 (3), 796–807. <https://doi.org/10.1152/jn.00614.2017>.
- Berridge, C.W., Schmeichel, B.E., España, R.A., 2012. Noradrenergic modulation of wakefulness/arousal. *Sleep. Med. Rev.* 16 (2), 187–197. <https://doi.org/10.1016/j.smrv.2011.12.003>.
- Blier, P., 2001. Crosstalk between the norepinephrine and serotonin systems and its role in the antidepressant response. *J. Psychiatry Neurosci.* JPN 26, S3–S10 (Suppl (Suppl)).
- Blusztajn, J.K., Wurtman, R.J., 1983. Choline and cholinergic neurons. *Sci. (N. Y., N. Y.)* 221 (4611), 614–620. <https://doi.org/10.1126/science.6867732>.
- Bourne, J.A., 2003. Intracerebral microdialysis: 30 years as a tool for the neuroscientist. *Clin. Exp. Pharmacol. Physiol.* 30 (1–2), 16–24. <https://doi.org/10.1046/j.1440-1681.2003.03789.x>.
- Bradberry, C.W., Barrett-Larimore, R.L., Jatlow, P., Rubino, S.R., 2000. Impact of self-administered cocaine and cocaine cues on extracellular dopamine in mesolimbic and sensorimotor striatum in rhesus monkeys. *J. Neurosci.* 20 (10), 3874–3883. <https://doi.org/10.1523/JNEUROSCI.20-10-03874.2000>.
- Calapai, A., Berger, M., Niessing, M., Heisig, K., Brockhausen, R., Treue, S., Gail, A., 2017. A cage-based training, cognitive testing and enrichment system optimized for rhesus macaques in neuroscience research. *Behav. Res. Methods* 49 (1), 35–45. <https://doi.org/10.3758/s13428-016-0707-3>.
- Cassidy, L., Treue, S., Gail, A., Pfeiffer, D., 2024. Choice-based severity scale (CSS): assessing the relative severity of procedures from a laboratory animal's perspective. *PeerJ* 12, e17300. <https://doi.org/10.7717/peerj.17300>.
- Centeno, M.L., Reddy, A.P., Smith, L.J., Sanchez, R.L., Henderson, J.A., Salli, N.C., Hess, D.J., Pau, F.K., Bethea, C.L., 2007. Serotonin in microdialysate from the mediobasal hypothalamus increases after progesterone administration to estrogen primed macaques. *Eur. J. Pharmacol.* 555 (1), 67–75. <https://doi.org/10.1016/j.ejphar.2006.10.027>.
- Chefer, V.I., Thompson, A.C., Zapata, A., Shippenberg, T.S., 2009. Overview of brain microdialysis. *Chapter 7, Unit 7.1 Curr. Protoc. Neurosci.* <https://doi.org/10.1002/0471142301.ns0701s47>.
- Chen, K.C., 2006. Effects of tissue trauma on the characteristics of microdialysis zero-net-flux method sampling neurotransmitters. *J. Theor. Biol.* 238 (4), 863–881. <https://doi.org/10.1016/j.jtbi.2005.06.035>.
- Chu, N., Bloom, F.E., 1973. Norepinephrine-containing neurons: changes in spontaneous discharge patterns during sleeping and waking. *Sci. (N. Y., N. Y.)* 179 (4076), 908–910. <https://doi.org/10.1126/science.179.4076.908>.
- Cools, R., Arnsten, A.F.T., 2022. Neuromodulation of prefrontal cortex cognitive function in primates: the powerful roles of monoamines and acetylcholine. *Neuropsychopharmacology* 47 (1), 309–328. <https://doi.org/10.1038/s41386-021-01100-8>.
- Crayen, M.A., Kagan, I., Esghaei, M., Hoehl, D., Thomas, U., Prückl, R., Schaffelhofer, S., Treue, S., 2024. Using camera-guided electrode microdrive navigation for precise 3D targeting of macaque brain sites. *PLoS One* 19 (5), e0301849. <https://doi.org/10.1371/journal.pone.0301849>.
- Dalley, J.W., McGaughy, J., O'Connell, M.T., Cardinal, R.N., Levita, L., Robbins, T.W., 2001. Distinct changes in cortical acetylcholine and noradrenaline efflux during contingent and noncontingent performance of a visual attentional task. *J. Neurosci.* 21 (13), 4908–4914. <https://doi.org/10.1523/JNEUROSCI.21-13-04908.2001>.
- Darvesh, A.S., Carroll, R.T., Geldenhuys, W.J., Gudelsky, G.A., Klein, J., Meshul, C.K., Van Der Schyf, C.J., 2011a. In vivo brain microdialysis: Advances in neuropsychopharmacology and drug discovery. *Expert Opin. Drug Discov.* 6 (2), 109–127. <https://doi.org/10.1517/17460441.2011.547189>.
- De Leo, G., Gulino, R., Coradazzi, M., Leanza, G., 2023. Acetylcholine and noradrenaline differentially regulate hippocampus-dependent spatial learning and memory. *Brain Commun.* 5 (1). <https://doi.org/10.1093/braincomms/fcac338>.
- Di Chiara, G., Tanda, G., Carboni, E., 1996. Estimation of in-vivo neurotransmitter release by brain microdialysis: the issue of validity. *Behav. Pharmacol.* 7 (7), 640–657.
- Di Giovanni, G., Di Matteo, V., Esposito, E., 2008. Serotonin-Dopamine Interaction. *Exp. Evid. Ther. Relev.* 172, 1–665.
- Engel, S., Fleischhauer, M., Lesch, K.P., Reif, A., Strobel, A., 2011. Serotonergic modulation in executive functioning: linking genetic variations to working memory performance. *Neuropsychologia* 49 (13), 3776–3785. <https://doi.org/10.1016/j.neuropsychologia.2011.09.038>.
- Folgering, J.H., Choi, M., Schlumbohm, C., van Gaalen, M.M., Stratford, R.E., 2019. Development of a non-human primate model to support CNS translational research: Demonstration with D-amphetamine exposure and dopamine response. *J. Neurosci. Methods* 317, 71–81. <https://doi.org/10.1016/j.jneumeth.2019.02.005>.
- Fumero, B., Guadalupe, T., Valladares, F., Mora, F., O'Neill, R.D., Mas, M., Gonzalez-Mora, J.L., 1994. Fixed versus removable microdialysis probes for in vivo neurochemical analysis: Implications for behavioral studies. *J. Neurochem.* 63 (4), 1407–1415. <https://doi.org/10.1046/j.1471-4159.1994.63041407>.
- Georgieva, J., Luthman, J., Mohring, B., Magnusson, O., Luthman, J., Mohring, B., 1993. Tissue and Microdialysate Changes After Repeated and Permanent Probe Implantation in the Striatum of Freely Moving Rats. *Brain Res. Bull.* 31 (5), 463–470. [https://doi.org/10.1016/0361-9230\(93\)90110-W](https://doi.org/10.1016/0361-9230(93)90110-W).
- Gold, P.E., 2014. Regulation of memory - From the adrenal medulla to liver to astrocytes to neurons. *Brain Res. Bull.* 105, 25–35. <https://doi.org/10.1016/j.brainresbull.2013.12.012>.
- Harris, K.D., Thiele, A., 2011. Cortical state and attention. *Nat. Rev. Neurosci.* 12 (9), 509–523. <https://doi.org/10.1038/nrn3084>.
- Hermenegildo, C., Monfort, P., Felipe, V., 2000. Activation of N-methyl-D-aspartate receptors in rat brain in vivo following acute ammonia intoxication: Characterization by in vivo brain microdialysis. *Hepatology* 31 (3), 709–715. <https://doi.org/10.1002/hep.510310322>.

- Hüer, J., Saxena, P., Treue, S., 2024. Pathway-selective optogenetics reveals the functional anatomy of top-down attentional modulation in the macaque visual cortex. *Proc. Natl. Acad. Sci. USA* 121 (3). <https://doi.org/10.1073/pnas.2304511121>.
- Huff, J.K., Davies, M.I., 2002. Microdialysis monitoring of methylphenidate in blood and brain correlated with changes in dopamine and rat activity. *J. Pharm. Biomed. Anal.* 29 (5), 767–777. [https://doi.org/10.1016/S0731-7085\(02\)00196-6](https://doi.org/10.1016/S0731-7085(02)00196-6).
- Jaquins-Gerstl, A., Michael, A.C., 2009. Comparison of the brain penetration injury associated with microdialysis and voltammetry. *J. Neurosci. Methods* 183 (2), 127–135. <https://doi.org/10.1016/j.jneumeth.2009.06.023>.
- Klinkenberg, I., Sambeth, A., Blokland, A., 2011. Acetylcholine and attention. *Behav. Brain Res.* 221 (2), 430–442. <https://doi.org/10.1016/j.bbr.2010.11.033>.
- Kozyrev, V., Daliri, M.R., Schwedhelm, P., Treue, S., 2019. Strategic deployment of feature-based attentional gain in primate visual cortex. *PLoS Biol.* 17 (8). <https://doi.org/10.1371/journal.pbio.3000387>.
- Krebs, R.M., Boehler, C.N., Roberts, K.C., Song, A.W., Woldorff, M.G., 2012. The involvement of the dopaminergic midbrain and cortico-striatal-thalamic circuits in the integration of reward prospect and attentional task demands. *Cereb. Cortex* 22 (3), 607–615. <https://doi.org/10.1093/cercor/bhr134>.
- Lee, S.H., Dan, Y., 2012. Neuromodulation of Brain States. *Neuron* 76 (1), 209. <https://doi.org/10.1016/J.NEURON.2012.09.012>.
- Martin-Fardon, R., Sandillon, F., Thibault, J., Privat, A., Vignon, J., 1997. Long-term monitoring of extracellular dopamine concentration in the rat striatum by a repeated microdialysis procedure. *J. Neurosci. Methods* 72 (2), 123–135. [https://doi.org/10.1016/S0165-0270\(96\)02170-X](https://doi.org/10.1016/S0165-0270(96)02170-X).
- McGaughy, J., Ross, R.S., Eichenbaum, H., 2008. Noradrenergic, but not cholinergic, deafferentation of prefrontal cortex impairs attentional set-shifting. *Neuroscience* 153 (1), 63–71. <https://doi.org/10.1016/j.neuroscience.2008.01.064>.
- McGinley, M.J., Vinck, M., Reimer, J., Batista-Brito, R., Zagher, E., Cadwell, C.R., Tolia, A.S., Cardin, J.A., McCormick, D.A., 2015. Waking State: Rapid Variations Modulate Neural and Behavioral Responses. *Neuron* 87 (6), 1143–1161. <https://doi.org/10.1016/J.NEURON.2015.09.012>.
- Megias, M., Emri, Z., Freund, T.F., Gulyás, A.I., 2001. Total number and distribution of inhibitory and excitatory synapses on hippocampal CA1 pyramidal cells. *Neuroscience* 102 (3), 527–540. [https://doi.org/10.1016/S0306-4522\(00\)00496-6](https://doi.org/10.1016/S0306-4522(00)00496-6).
- Millichap, J.G., 1996. Catecholamines in attention deficit disorders. *Pediatr. Neurol.* Briefs 10 (3), 17–18. <https://doi.org/10.15844/pedneurbriefs-10-3-1>.
- Mitchell, H.A., Weinschenker, D., 2010. Good night and good luck: Norepinephrine in sleep pharmacology. *Biochem. Pharmacol.* 79 (6), 801–809. <https://doi.org/10.1016/j.bcp.2009.10.004>.
- Nandi, P., Lunte, S.M., 2009. Recent trends in microdialysis sampling integrated with conventional and microanalytical systems for monitoring biological events: a review. *Anal. Chim. Acta* 651 (1), 1–14. <https://doi.org/10.1016/j.aca.2009.07.064>.
- Nesbitt, K.M., Jaquins-Gerstl, A., Skoda, E.M., Wipf, P., Michael, A.C., 2013. Pharmacological mitigation of tissue damage during brain microdialysis. *Anal. Chem.* 85 (17), 8173–8179. <https://doi.org/10.1021/ac401201x>.
- Petroff, O.A.C., 2002. GABA and glutamate in the human brain. *Neurosci.: A Rev. J. Bringing Neurobiol., Neurol. Psychiatry* 8 (6), 562–573. <https://doi.org/10.1177/1073858402238515>.
- Pfefferle, D., Plümer, S., Burchardt, L., Treue, S., Gail, A., 2018. Assessment of stress responses in rhesus macaques (*Macaca mulatta*) to daily routine procedures in system neuroscience based on salivary cortisol concentrations. *PLoS One* 13 (1), e0190190. <https://doi.org/10.1371/journal.pone.0190190>.
- Pisani, A., Bonsi, P., Centonze, D., Calabresi, P., Bernardi, G., 2000. Activation of D2-like dopamine receptors reduces synaptic inputs to striatal cholinergic interneurons. *J. Neurosci.* 20 (7). <https://doi.org/10.1523/JNEUROSCI.20-07-J0003.2000>.
- Pratt, K.G., Aizenman, C.D., 2007. Homeostatic regulation of intrinsic excitability and synaptic transmission in a developing visual circuit. *J. Neurosci.* 27 (31), 8268–8277. <https://doi.org/10.1523/JNEUROSCI.1738-07.2007>.
- Roach, C.J. (2023). *Investigating the Origin and Role of Catecholamines in the Visual Cortex of the Macaque Monkey*. Dissertation, Duke University. Retrieved from (<https://hdl.handle.net/10161/27609>).
- Roberts, M.J., Zinke, W., Guo, K., Robertson, R., McDonald, J.S., Thiele, A., 2005. Acetylcholine dynamically controls spatial integration in marmoset primary visual cortex. *J. Neurophysiol.* 93 (4), 2062–2072. <https://doi.org/10.1152/jn.00911.2004>.
- Roelfsema, P.R., Treue, S., 2014. Basic neuroscience research with nonhuman primates: A small but indispensable component of biomedical research. *Neuron* 82 (6), 1200–1204. <https://doi.org/10.1016/j.neuron.2014.06.003>.
- Sarter, M., Paolone, G., 2011. Deficits in Attentional Control: Cholinergic Mechanisms and Circuitry-Based Treatment Approaches. *Behav. Neurosci.* 125 (6), 825–835. <https://doi.org/10.1037/a0026227>.
- Schuerman, W.L., Chandrasekaran, B., Leonard, M.K., 2022. Arousal States as a Key Source of Variability in Speech Perception and Learning. *Languages* 2022 7 (1), 19. <https://doi.org/10.3390/LANGUAGES7010019>.
- Schultz, W., 2013. Updating dopamine reward signals. *Curr. Opin. Neurobiol.* 23 (2), 229–238. <https://doi.org/10.1016/j.conb.2012.11.012>.
- Schwedhelm, P., Baldauf, D., Treue, S., 2017. Electrical stimulation of macaque lateral prefrontal cortex modulates oculomotor behavior indicative of a disruption of top-down attention. *Sci. Rep.* 7 (1). <https://doi.org/10.1038/s41598-017-18153-9>.
- Thiollier, T., Wu, C., Porras, G., Bezard, E., Li, Q., Zhang, J., Contamin, H., 2018. Microdialysis in awake macaque monkeys for central nervous system pharmacokinetics. *Anim. Models Exp. Med.* 1 (4), 314–321. <https://doi.org/10.1002/ame2.12046>.
- Treue, S., Lemon, R., 2023. The indispensable contribution of nonhuman primates to biomedical research. In: Robinson, L.M., Weiss, A. (Eds.), *Nonhuman primate welfare: From history, science, and ethics to practice*. Springer Nature Switzerland AG, pp. 589–603. https://doi.org/10.1007/978-3-030-82708-3_24.
- Van den Brink, R.L., Pfeffer, T., Donner, T.H., 2019. Brainstem Modulation of Large-Scale Intrinsic Cortical Activity Correlations. In: *Frontiers in Human Neuroscience*, 13, p. 340. <https://doi.org/10.3389/fnhum.2019.00340>.
- Van Someren, E.J.W., 2021. Brain mechanisms of insomnia: new perspectives on causes and consequences. *Physiol. Rev.* 101 (3), 995–1046. <https://doi.org/10.1152/physrev.00046.2019>.
- Watson, C.J., Venton, B.J., Kennedy, R.T., 2006. In vivo measurements of neurotransmitters by microdialysis sampling. *Anal. Chem.* 78 (5), 1391–1399. <https://doi.org/10.1021/ac0693722>.
- Weiss, J.M., Goodman, P.A., Losito, B.G., Corrigan, S., Charry, J.M., Bailey, W.H., 1981. Behavioral depression produced by an uncontrollable stressor: Relationship to norepinephrine, dopamine, and serotonin levels in various regions of rat brain. *Brain Res. Rev.* 3 (2), 167–205. [https://doi.org/10.1016/0165-0173\(81\)90005-9](https://doi.org/10.1016/0165-0173(81)90005-9).
- Westerink, B.H., 1995. Brain microdialysis and its application for the study of animal behaviour. *Behav. Brain Res.* 70 (2), 103–124. [https://doi.org/10.1016/0166-4328\(95\)80001-8](https://doi.org/10.1016/0166-4328(95)80001-8).
- Westerink, B.H., De Vries, J.B., 1988. Characterization of in vivo dopamine release as determined by brain microdialysis after acute and subchronic implantations: methodological aspects. *J. Neurochem.* 51 (3), 683–687. <https://doi.org/10.1111/j.1471-4159.1988.tb01798.x>.
- Yao, T., Treue, S., Krishna, B.S., 2016. An Attention-Sensitive Memory Trace in Macaque MT Following Saccadic Eye Movements. *PLoS Biol.* 14 (2). <https://doi.org/10.1371/journal.pbio.1002390>.
- Young, A.M., Bradford, H.F., 1986. Excitatory amino acid neurotransmitters in the corticostriate pathway: studies using intracerebral microdialysis in vivo. *J. Neurochem.* 47 (5), 1399–1404. <https://doi.org/10.1111/j.1471-4159.1986.tb00771.x>.
- Zamboni, W.C., 2007. Carrier-mediated and artificial cell-targeted cancer drug delivery. *Artif. Cells, Cell Eng. Ther.* 469–501. <https://doi.org/10.1533/9781845693077.4.469>.
- Zhang, M.Y., Beyer, C.E., 2006. Measurement of neurotransmitters from extracellular fluid in brain by in vivo microdialysis and chromatography-mass spectrometry. *J. Pharm. Biomed. Anal.* 40 (3), 492–499. <https://doi.org/10.1016/j.jpba.2005.07.025>.
- Zhou, S., Yu, Y., 2018. Synaptic E-I balance underlies efficient neural coding. *Front. Neurosci.* 12, 46. <https://doi.org/10.3389/fnins.2018.00046>.